

Complex Analysis Qualifying Exam Review

Fall 2026

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Chapter 1

Fall 2025

Problem 1.1 (Theorem 5.2, Stein, Chapter 2). Let $\{f_n\}$ be a sequence of holomorphic functions on the open unit disk

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\},$$

and suppose that $f_n \rightarrow f$ uniformly on compact subsets of $\mathbb{D}$, where $f : \mathbb{D} \rightarrow \mathbb{C}$ is continuous. Prove that f is holomorphic on $\mathbb{D}$.

Problem 1.2. 6 Use contour integration to evaluate

$$\int_0^\infty \frac{x^2}{x^4 + 1} dx.$$

Problem 1.3. 7 Let f be an analytic function on the open unit disk

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\},$$

and suppose that

$$|f(z)| \geq \sqrt{|z|}$$

for all $z \in \mathbb{D}$. Show that

$$|f(z)| \geq 1 \quad \text{for all } z \in \mathbb{D}.$$

Problem 1.4. 8 Determine how many solutions

$$z^7 - 7z^4 + z^3 - 3z = 1$$

are in the annulus

$$1 < |z| < 2.$$

Chapter 2

Cauchy's integral formula

Proposition 2.1 (Cauchy's theorem). Let $f : U \rightarrow \mathbb{C}$ be holomorphic in U , if γ is a closed smooth curve whose interior is contained in U , then

$$\int_{\gamma} f(z) dz = 0$$

Proposition 2.2 (Cauchy's integral formula). Let $f : U \rightarrow \mathbb{C}$ be holomorphic in U which contains the closure of a disk $D_r(z_0)$, for some $z_0 \in U$. Then for any $z \in D_r(z_0)$,

$$f(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{w - z} dw$$

where $\gamma = z_0 + re^{i\theta}$, $\theta \in [0, 2\pi]$. Moreover,

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_{\gamma} \frac{f(w)}{(w - z)^{n+1}} dw$$

Proposition 2.3 (Cauchy's inequality). Let $f : U \rightarrow \mathbb{C}$ be holomorphic in U which contains the closure of a disk $D_r(z_0)$, for some $z_0 \in U$. Then

$$|f^{(n)}(z_0)| \leq \frac{n! \sup_{|z|=r} |f(z)|}{r^n}$$

Proposition 2.4 (Liouville's theorem). Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be a bounded entire function, then f must be constant.

Proposition 2.5 (Holomorphic and analytic). Let $f : U \rightarrow \mathbb{C}$ be holomorphic in U which contains the closure of a disk $D_r(z_0)$, for some $z_0 \in U$. Then f has a power series expansion at z_0 :

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

for all $z \in D_r(z_0)$, where

$$a_n = \frac{f^{(n)}(z_0)}{n!}$$

This combined with the previous theorem gives a bound on the coefficients:

$$|a_n| \leq \frac{\sup_{|z|=r} |f(z)|}{r^n}$$

Proposition 2.6 (Morera's theorem). Suppose f is a continuous function in an open set U , if for any triangle T contained in D ,

$$\int_T f(z) dz = 0$$

then f is holomorphic.

Problem 2.1. Evaluate the following integral:

$$\int_0^{\infty} \frac{1 - \cos x}{x^2} dx$$

Problem 2.2. Let U be a region. Suppose f is holomorphic in U , and a sequence $\{z_n\} \subset U$ converges to some $z \in U$. If $f(z_n) = 0$ for all n , show that $f \equiv 0$.

Problem 2.3 (Spring 2018, Q4). Let f be an entire function, suppose that for each z_0 , the power series expansion

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

has at least one coefficient $a_n = 0$. Show that f must be a polynomial.

Problem 2.4 (Fall 2019, Q2). Let f be an entire function such that

$$\sup_{|z|>R} |f(z)| \leq AR^k + B$$

for positive constants A, B and all R large. Then show f is a polynomial with degree at most k .

Problem 2.5 (Fall 2023, Q1). (a) Show that if an entire function f satisfies

$$|f(z)| \leq \sqrt{|z|}$$

for all $z \in \mathbb{C}$, then $f(z)$ is a constant function.

(b) Why isn't the function $z^{1/2}$ a counterexample for Part (a)?

Problem 2.6 (Fall 2018, Q4). Suppose f, g are both entire functions such that $|f(z)| \leq |g(z)|$ for all $z \in \mathbb{C}$. Show that there exists $c \in \mathbb{C}$ such that $f = cg$.

Problem 2.7 (Spring 2015, Q6). Suppose f is a nonconstant entire function, then show that $f(\mathbb{C})$ is dense in $\mathbb{C}$.

Chapter 3

Singularities

Definition 3.1 (residue). If f has a pole of order n at z_0 , then

$$\operatorname{res}_{z_0} f = \lim_{z \rightarrow z_0} \frac{1}{(n-1)!} \left(\frac{d}{dz} \right)^{n-1} (z - z_0)^n f(z)$$

Proposition 3.1 (Removable singularity). Let f be a holomorphic function defined on $D_r(z_0) \setminus \{z_0\}$. We say f has a removable singularity at z_0 iff

$$\lim_{z \rightarrow z_0} (z - z_0) f(z) = 0$$

iff there exists a neighborhood N of z_0 where f is bounded on N .

Proposition 3.2 (Pole). Let f be a holomorphic function defined on $D_r(z_0) \setminus \{z_0\}$. We say f has a pole at z_0 iff

$$\lim_{z \rightarrow z_0} |f(z)| = \infty$$

iff there exists a non-vanishing holomorphic function h and a unique n such that

$$f(z) = (z - z_0)^{-n} h(z)$$

Proposition 3.3 (Residue theorem). Suppose f is holomorphic in an open set containing a smooth closed curve γ and its interior, except for poles at points $z_1, \dots, z_n$ inside of γ . Then

$$\int_{\gamma} f(z) dz = 2\pi i \sum_{i=1}^n \operatorname{res}_{z_i} f$$

Proposition 3.4 (Casorati-Weierstrass). Let f be a holomorphic function defined on $D_r(z_0) \setminus \{z_0\}$ and suppose f has an essential singularity at z_0 , then the image of f is dense in $\mathbb{C}$.

Proposition 3.5 (Argument principle). Let f be a meromorphic function in an open set containing a smooth closed curve γ and its interior. If f has no poles and never vanishes on γ , then

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} = (\#\text{zeros of } f \text{ inside } \gamma) - (\#\text{poles of } f \text{ inside } \gamma)$$

where zeros and poles are counted with multiplicities.

Proposition 3.6 (Rouche's theorem). Let f be a meromorphic function in an open set containing a smooth closed curve γ and its interior. If

$$|g(z)| < |f(z)| \text{ for all } z \in \gamma$$

then $f, f + g$ have the same number of zeros inside of γ .

Proposition 3.7 (Open mapping theorem). If f is holomorphic and non-constant in a region Ω , then f is open.

Proposition 3.8 (Maximum modulus principle). If f is holomorphic on a region Ω with compact closure $\bar{\Omega}$ and continuous on $\bar{\Omega}$, then

$$\sup_{z \in \Omega} |f(z)| \leq \sup_{z \in \partial\Omega} |f(z)|$$

(There are many equivalent statements of MMP, the idea is the same: a nonconstant holomorphic function f cannot attain the maximum in the interior of the compact set).

Proposition 3.9 (Mean-value property). If f is holomorphic in a disc $D_R(z_0)$, then

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta}) d\theta, \quad \text{for any } 0 < r < R.$$

Problem 3.1. Compute

$$\int_0^{\infty} \frac{1}{1+x^2} dx$$

Problem 3.2. Compute the following integral: fix $0 < a < 1$,

$$\int_{-\infty}^{\infty} \frac{e^{ax}}{1+e^x} dx$$

Problem 3.3. Assume that $f : \mathbb{C} \rightarrow \mathbb{C}$ is an entire function, not identically equal to 0, and let

$$Z = \{z \in \mathbb{C} : f(z) = 0\}.$$

Prove that if Z is unbounded, then f has an essential singularity at ∞ .

Problem 3.4. Suppose f is an injective entire function, show f is linear.

Problem 3.5 (Spring 2017, Q6). Let $f : U \rightarrow \mathbb{C}$ be a non-constant holomorphic function where $U \subset \mathbb{C}$ is an open set containing the closure $\overline{\mathbb{D}}$ of the unit disk

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}.$$

If $|f(z)| = 1$ for all $z \in \partial\mathbb{D}$, then prove that

$$\mathbb{D} \subset f(\overline{\mathbb{D}}).$$

Problem 3.6 (Fall 2019, Q4). Let f be a holomorphic function in the punctured disk

$$\{z : 0 < |z| < 2\}$$

satisfying

$$|f(z)| \leq \left(\log \frac{1}{|z|}\right)^{100} \quad \text{in } \{z : |z| \leq 1/2\},$$

and

$$|f(z)| = 1 \quad \text{on } |z| = 1.$$

- (a) Show that f has a removable singularity at the origin.
- (b) Show that if $f(z) \neq 0$ in $|z| < 1$, then f is constant.

Chapter 4

Conformal mapping

Definition 4.1 (automorphisms of the disk). Let $a \in \mathbb{D}$, the following function is an automorphism of the disk:

$$\phi_a(z) = \frac{a - z}{1 - \bar{a}z}$$

We have $\phi_a \circ \phi_a = \text{id}$. Moreover, if f is a holomorphic function on $\mathbb{D}$, then $g(0) = 0$ if we let

$$g = \phi_{f(a)} \circ f \circ \phi_a$$

Proposition 4.1 (Schwarz lemma). Let $f : \mathbb{D} \rightarrow \mathbb{D}$ be holomorphic with $f(0) = 0$, then

1. $|f(z)| \leq |z|$ for all $z \in \mathbb{D}$.
2. If $|f(z_0)| = |z_0|$ for some $z_0 \neq 0$, then $f(z) = cz$ for some $|c| = 1$.
3. $|f'(0)| \leq 1$ and if equality holds, then $f(z) = cz$ for some $|c| = 1$.

Proposition 4.2 (Corollary of Schwarz's lemma). If f is an automorphism of $\mathbb{D}$, then there exist $\theta \in \mathbb{R}$ and $a \in \mathbb{D}$ such that

$$f(z) = e^{i\theta} \frac{a - z}{1 - \bar{a}z}$$

Namely, the only automorphisms of $\mathbb{D}$ that fix the origin are rotations.

Proposition 4.3 (Automorphisms of $\mathbb{H}$). f is an automorphism of $\mathbb{H}$ iff it is of the form

$$f(z) = \frac{az + b}{cz + d}$$

where $a, b, c, d \in \mathbb{R}$, $ad - bc = 1$.

Problem 4.1 (Spring 2019, Q4). Show that the punctured unit disk

$$\{z : 0 < |z| < 1\}$$

and the annulus

$$\{z : 1 < |z| < 2\}$$

cannot be conformally equivalent.

Problem 4.2 (Fall 2019, Q4). Let $f(z)$ be a holomorphic function on

$$\mathbb{D} := \{z \in \mathbb{C} : |z| < 1\},$$

with $|f(z)| < 1$, and suppose $f(\alpha) = 0$ for some $|\alpha| < 1$. Show that for $z \in \mathbb{D}$,

$$|f(z)| \leq \left| \frac{z - \alpha}{1 - \bar{\alpha}z} \right|.$$

Problem 4.3 (Spring 2016, Q5). Let $f : \mathbb{D} \rightarrow \mathbb{D}$ be a holomorphic function where

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$$

is the unit disk. Prove that if f has at least 2 fixed points then f is the identity map.